Differential Dynamic Programming Formulations for Multi-Period Optimal Power Flow

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I. SOCP MPOPF FORMULATION

Let $\mathcal{N}$, $\mathcal{L}$, $\mathcal{B}$, and $\mathcal{D}$ denote buses, directed branches, battery buses, and PV buses; $j_{\text{subs}} = 1$ is the substation bus and $\mathcal{C}(j)$ contains the children of j. The network MPOPF is

$$
\begin{aligned}
\min \quad & \sum_{t\in\mathcal{T}} c^t S_{\text{base}} P^t_{\text{Subs}} \Delta t \\
& + \sum_{t\in\mathcal{T}} \sum_{j\in\mathcal{B}} C_B \left(S_{\text{base}} P^t_{B,j}\right)^2 \Delta t \\
& + w \sum_{j\in\mathcal{B}} \left(B^T_j - B_{\text{init},j}\right)^2 && (1)\\
\text{s.t.} \quad & P^t_{\text{Subs}} - \sum_{(j_{\text{subs}},k)\in\mathcal{L}} P^t_{j_{\text{subs}}k} = 0, && (2)\\
& P^t_{\text{Subs}} \ge 0, && (3)\\
& Q^t_{\text{Subs}} - \sum_{(j_{\text{subs}},k)\in\mathcal{L}} Q^t_{j_{\text{subs}}k} = 0, && (4)\\
& \sum_{k\in\mathcal{C}(j)} P^t_{jk} - P^t_{ij} + r_{ij}\ell^t_{ij} = P^t_{B,j} + p^t_{D,j} - p^t_{L,j}, && (5)\\
& \sum_{k\in\mathcal{C}(j)} Q^t_{jk} - Q^t_{ij} + x_{ij}\ell^t_{ij} = q^t_{D,j} - q^t_{L,j}, && (6)\\
& v^t_j = v^t_i - 2(r_{ij}P^t_{ij} + x_{ij}Q^t_{ij}) + (r^2_{ij} + x^2_{ij})\ell^t_{ij}, && (7)\\
& (P^t_{ij})^2 + (Q^t_{ij})^2 \le v^t_i \ell^t_{ij}, && (8)\\
& \ell^t_{ij} \ge 0, && (9)\\
& v^t_{\text{Subs}} = 1.05^2, && (10)\\
& \underline{V}^2_j \le v^t_j \le \overline{V}^2_j, && (11)\\
& -q^{\max,t}_{D,j} \le q^t_{D,j} \le q^{\max,t}_{D,j}, && (12)\\
& q^{\max,t}_{D,j} = \sqrt{S^2_{D,j} - (p^t_{D,j})^2}, && (13)\\
& B^t_j = B^{t-1}_j - P^t_{B,j}\Delta t, && (14)\\
& B^0_j = B_{\text{init},j}, && (15)\\
& \text{soc}^{\min}_j B_{R,j} \le B^t_j \le \text{soc}^{\max}_j B_{R,j}, && (16)\\
& -P^R_{B,j} \le P^t_{B,j} \le P^R_{B,j}. && (17)
\end{aligned}
$$

II. IMPLEMENTATION AND RESULTS

The network transcription uses analytic derivatives and accepts time-varying load, PV, and price data directly at each stage. Compact problems retain FilterDDP's dense null-space backward pass. For feeder models, sparse Jacobians and local Hessian blocks are instead assembled into

$$
\begin{bmatrix} \underset{n_u\times n_u}{H} & \underset{n_u\times n_c}{A^\mathsf{T}} \\ \underset{n_c\times n_u}{A} & \underset{n_c\times n_c}{0} \end{bmatrix}
\begin{bmatrix} \underset{n_u\times 1}{\Delta u} \\ \underset{n_c\times 1}{\Delta\lambda} \end{bmatrix}
= \underset{(n_u+n_c)\times 1}{b}, \tag{18}
$$

For a stage with n_u controls and n_c equality constraints, $H \in \mathbb{R}^{n_u\times n_u}$, $A \in \mathbb{R}^{n_c\times n_u}$, $\Delta u \in \mathbb{R}^{n_u}$, $\Delta\lambda \in \mathbb{R}^{n_c}$, and $b \in \mathbb{R}^{n_u+n_c}$; hence the full block matrix is $(n_u+n_c)\times(n_u+n_c)$. At a high level, H collects the local curvature terms used by the backward step, A is the linearized equality-constraint Jacobian, and Δu and $\Delta\lambda$ are the control and equality-multiplier updates; b collects the corresponding first-order residual terms. A complete interpretation of these blocks requires the FilterDDP derivation and is beyond the present working note. The system is solved by sparse LU without explicitly constructing a dense QR basis or reduced Hessian. The dense route has less overhead for small problems; the sparse route is required when those dense intermediate matrices become impractical. Reported times include FilterDDP optimization iterations, not model construction. Every successful network result used the complete loss-aware SOCP model and was checked against an independently validated centralized Ipopt or Gurobi solution.

TABLE I
FILTERDDP NETWORK TEST SUMMARY. A CHECK DENOTES AGREEMENT WITH THE CENTRALIZED SOLUTION AT EVERY LISTED HORIZON; IT DOES NOT IMPLY COMPETITIVE RUNTIME.

System	Solved	Tested horizons T
IEEE123C	✓	3, 6, 12, 24, 48, 96
IEEE2522C	✓	3, 6, 12, 24
large10kC	×	3

TABLE II
SUCCESSFUL FILTERDDP NETWORK SOLVES

System	T	Iterations	Solve time (s)
IEEE123C	3	48	13.856
	6	60	143.647
	12	81	306.817
	24	97	610.766
	48	111	923.073
	96	121	2021.412
IEEE2522C	3	56	233.098
	6	82	836.933
	12	79	1902.103
	24	84	3068.228

IEEE123C has per-stage dimensions $(n_x, n_u, n_c) = (51, 791, 562)$, while IEEE2522C has $(250, 13358, 10337)$. The original dense IEEE2522C backward pass could not complete one iteration because it attempted a dense 13358×13358 QR basis and a reduced Hessian of order 3021. The sparse KKT route removed that storage bottleneck and reached a realistic $T = 24$ horizon, but required 51.1 min; the corresponding centralized Gurobi solver time was 27.2 s. Reusing UMFPACK's symbolic factorization did not help: the identical $T = 3$ solution slowed from 233.1 to 295.3 s. Thus the present result establishes correctness and memory feasibility, not computational competitiveness.

For large10kC at $T = 3$, the per-stage dimensions are $(n_x, n_u, n_c) = (1020, 54665, 42303)$. Blocking the multiple right-hand-side KKT solve avoided the earlier immediate allocation failure. However, the run produced no terminal iteration report after two hours and was stopped manually. No FilterDDP convergence or timing claim is therefore made for this system. Appendix B records the test-data corrections made before the IEEE2522C and large10kC comparisons.

III. Current Agenda

The immediate agenda is to understand equality-constrained DDP from first principles, using the treatment of El Kazdadi *et al.* [1] (reference [21] in the FilterDDP paper [2]) as a guide. The purpose is to identify the power-flow-specific dynamics, constraints, and derivative expressions directly, and to develop a reliable working understanding of the complete algorithm before making further formulation or scalability claims.

Appendix A
First-Order DDP Derivation

The copper-plate problem below is included as the reference problem used to derive and test the first-order Differential Dynamic Programming workflow. It keeps the temporal battery coupling explicit without introducing network equations, so the information passed between period subproblems can be seen directly.

A. Copper-Plate Reference Problem

A storage device and grid connection serve demand p_L^t over $\mathcal{T} = \{1,\dots,T\}$. Here $P_B^t > 0$ discharges, and B^t is the end-of-period energy; hence B^0 is initial and B^T is terminal. The sign convention follows [3].

$$\begin{aligned}
\min_{P_{\text{Subs}},\, P_B,\, B} \quad & \sum_{t=1}^{T}\Big[c^t P_{\text{Subs}}^t + C_B\big(P_B^t\big)^2\Big]\Delta t \\
& \quad + \; w\big(B^T - B_{\text{init}}\big)^2 && \text{(19a)}\\
\text{s.t.} \quad & P_{\text{Subs}}^t + P_B^t - p_L^t = 0, \qquad t\in\mathcal{T} && \text{(19b)}\\
& B^0 = B_{\text{init}} && \text{(19c)}\\
& B^t = B^{t-1} - P_B^t \Delta t, \qquad t\in\mathcal{T} && \text{(19d)}\\
& P_{\text{Subs}}^t \ge 0, \qquad t\in\mathcal{T} && \text{(19e)}\\
& \underline{P_B} \le P_B^t \le \overline{P_B}, \qquad t\in\mathcal{T} && \text{(19f)}\\
& \underline{B} \le B^t \le \overline{B}, \qquad t\in\mathcal{T} && \text{(19g)}
\end{aligned}$$

The *base instance* is (19a)–(19d), with all three bounds dropped.

B. Reference Instance Data

TABLE III
Copper-Plate Instance Data

Interval t	1	2	3	4	5	6
Demand p_L^t	1.657	1.888	1.998	1.834	1.623	1.657
Energy price c^t	0.140	0.197	0.175	0.105	0.083	0.140

Period length Δt	1	Initial energy B_{init}	2.0
Throughput C_B	0.05	Terminal weight w	5.0

Bounds, constrained variants only	
$\underline{P_B}$, $\overline{P_B}$	$-0.45,\ 0.45$
$\underline{B}$, $\overline{B}$	$1.20,\ 1.95$

All quantities are in per unit on a 1000-kVA base, 1 p.u. = 1000 kW and 1 p.u.h = 1000 kWh. Bounds apply only to the constrained variants.

The derivation is stated directly in MPOPF notation. Equality multipliers are denoted by λ and inequality multipliers by μ. Thus the dynamics multiplier called $\mu[t]$ in the earlier derivation is written λ_{soc}^t here.

C. Centralized Lagrangian

Write every inequality as $g \le 0$. The Lagrangian of (19) is

$$\begin{aligned}
\mathcal{L} = & \sum_{t=1}^{T}\left[c^t P_{\text{Subs}}^t + C_B (P_B^t)^2\right]\Delta t + w(B^T - B_{\text{init}})^2 \\
& + \sum_{t=1}^{T} \lambda_{\text{bal}}^t (P_{\text{Subs}}^t + P_B^t - p_L^t) \\
& + \sum_{t=1}^{T} \lambda_{\text{soc}}^t (B^t - B^{t-1} + \Delta t P_B^t) \\
& + \sum_{t=1}^{T} \mu_{\underline{P_{\text{Subs}}}}^t (-P_{\text{Subs}}^t) \\
& + \sum_{t=1}^{T} \left[\mu_{\underline{P_B}}^t (\underline{P_B} - P_B^t) + \mu_{\overline{P_B}}^t (P_B^t - \overline{P_B})\right] \\
& + \sum_{t=1}^{T} \left[\mu_{\underline{B}}^t (\underline{B} - B^t) + \mu_{\overline{B}}^t (B^t - \overline{B})\right]. \qquad (20)
\end{aligned}$$

Inter-period optimality condition. Here $B^0 = B_{\text{init}}$ is fixed. The local SPOPF solver enforces primal and dual feasibility, complementary slackness, and stationarity with respect to its own power variables. The condition that couples adjacent periods is stationarity with respect to B^t, for $t = 1,\dots,T-1$:

$$0 = \lambda_{\text{soc}}^t - \lambda_{\text{soc}}^{t+1} - \mu_{\underline{B}}^t + \mu_{\overline{B}}^t, \qquad (21)$$

Equation (21) shows why the successor multiplier must enter the preceding period. At $t = T$, the terminal penalty already included in the final SPOPF terminates this chain.

D. Per-Period Decomposition

At sweep k, period t receives $\boldsymbol{B^{k,t-1}}$ from the period just solved and the successor multiplier $\boldsymbol{\lambda_{\mathrm{soc}}^{k-1,t+1}}$ from the previous sweep. It solves

$$\min_{P^t_{\mathrm{Subs}},P^t_B,B^t} \quad c^t P^t_{\mathrm{Subs}}\Delta t + C_B(P^t_B)^2\Delta t - \boldsymbol{\lambda_{\mathrm{soc}}^{k-1,t+1}} B^t \tag{22}$$

$$\text{s.t.} \quad P^t_{\mathrm{Subs}} + P^t_B = p^t_L, \tag{23}$$

$$B^t - \boldsymbol{B^{k,t-1}} + \Delta t P^t_B = 0, \tag{24}$$

$$P^t_{\mathrm{Subs}} \geq 0, \quad \underline{P_B} \leq P^t_B \leq \overline{P_B}, \quad \underline{B} \leq B^t \leq \overline{B}. \tag{25}$$

For $t = T$, the last term of (22) is replaced by $w(B^T - B_{\mathrm{init}})^2$. After solving, the equality multiplier is read from the local SOC equation,

$$\lambda^{k,t}_{\mathrm{soc}} = -\,\mathrm{dual}\big(B^t - \boldsymbol{B^{k,t-1}} + \Delta t P^t_B = 0\big)\,. \tag{26}$$

The term $-\boldsymbol{\lambda_{\mathrm{soc}}^{k-1,t+1}} B^t$ inserts the successor term missing from a standalone OPF's stationarity in B^t. The earlier program wrote the equivalent expression

$$\boldsymbol{\lambda_{\mathrm{soc}}^{k-1,t+1}} \left(B^{k-1,t+1} - B^t + \Delta t P^{t+1}_B\right). \tag{27}$$

Because $B^{k-1,t+1}$ is fixed and the auxiliary P^{t+1}_B is not the next period's actual decision, only the $-\boldsymbol{\lambda_{\mathrm{soc}}^{k-1,t+1}} B^t$ part supplies the intended inter-period sensitivity.

E. Sweep

The workflow is

$$t = 1 \to T: \quad \boldsymbol{B^{k,t-1}} \to (22) \to (B^{k,t}, \lambda^{k,t}_{\mathrm{soc}}), \qquad k \leftarrow k{+}1, \tag{28}$$

and stops when

$$\max\big(\|B^k - B^{k-1}\|_2, \|\lambda^k_{\mathrm{soc}} - \lambda^{k-1}_{\mathrm{soc}}\|_2\big) < \epsilon. \tag{29}$$

This passes only the first derivative of the future cost. Filter-DDP also passes its curvature V^t_{BB}; the first-order workflow has no corresponding quantity.

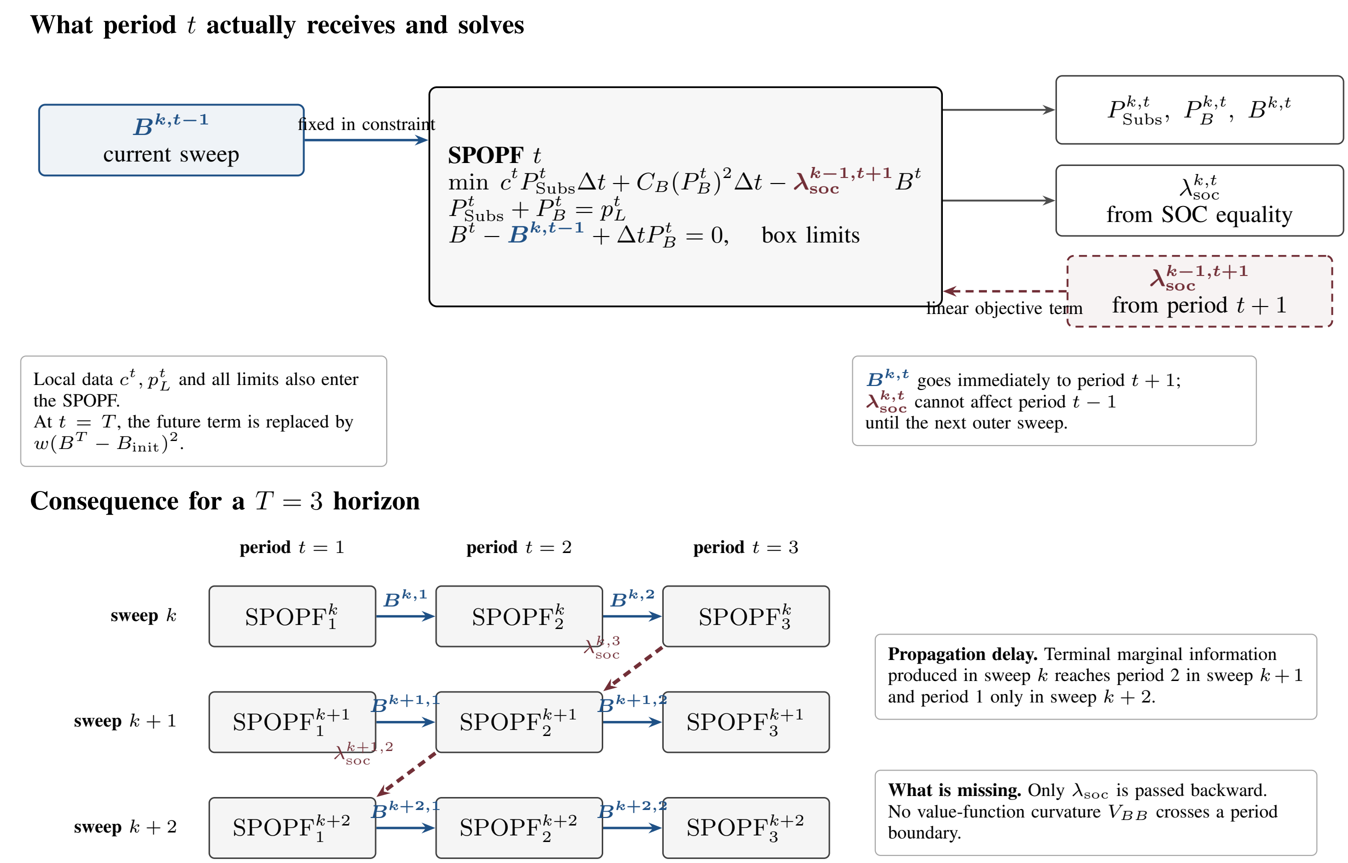

Fig. 1. Information flow in the first-order DDP sweep.

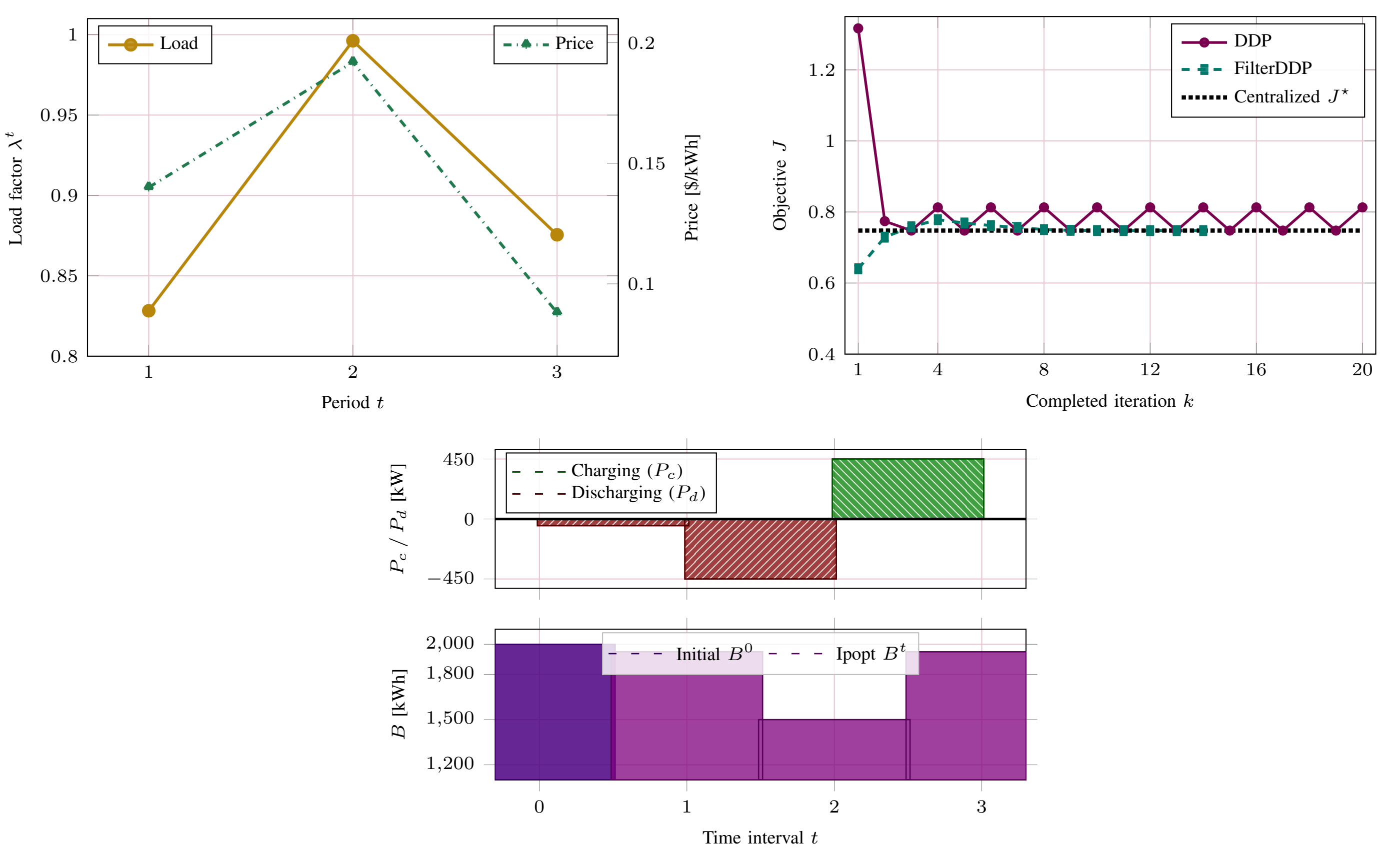

Fig. 2. Three-period inputs, objective trajectories, and optimal battery actions.

TABLE IV
SOC EQUALITY MULTIPLIERS DURING THE TWO-CYCLE

k	$\lambda_{\mathrm{soc}}^{k,1}$	$\lambda_{\mathrm{soc}}^{k,2}$	$\lambda_{\mathrm{soc}}^{k,3}$
15	0.135 00	0.088 04	0.314 29
16	0.088 04	0.231 96	0.088 04
17	0.135 00	0.088 04	0.314 29
18	0.088 04	0.231 96	0.088 04
19	0.135 00	0.088 04	0.314 29
20	0.088 04	0.231 96	0.088 04

TABLE V
IDENTICAL-INSTANCE SOLVER COMPARISON

Method	Outcome	Iterations	Objective J
Centralized Ipopt	optimal	—	0.747650349
FilterDDP	optimal	14	0.747650349
First-order DDP	two-cycle	> 2000	0.747650353/0.812950270

The undamped sweep does not converge. Through 2000 sweeps it remains in an exact period-two cycle: the residual in (29) is 0.602079, and the objective alternates between 0.74765035 and 0.81295027. The centralized objective is 0.747650349; on the other branch the battery dispatch differs by as much as 0.850 p.u. (850 kW). Thus even this three-period copper-plate problem exposes the instability of the first-order iteration. On the identical data, bounds, and initialization, FilterDDP terminates normally in 14 iterations. Its objective differs from centralized Ipopt by 1.48×10^{-10}, while its maximum battery-power and energy differences are 6.50×10^{-9} p.u. and 3.71×10^{-9} p.u.h, respectively. Consequently, the two-cycle is a property of the first-order DDP iteration, not of an unusual or ill-posed optimization instance.

APPENDIX B
NETWORK DATA CORRECTIONS

These corrections concern test-data conversion rather than the MPOPF model. The feeder formerly labelled "IEEE2552C" was renamed IEEE2522C. Its one-phase aggregate conversion combined aggregate three-phase powers with a phase-voltage base, making per-unit line impedances nine times too large. The electrical base was corrected to 7.2 kV on 1 MVA; physical impedances, powers, DER ratings, voltage limits, losses, and SOC constraints were unchanged.

For the synthetic large10kC feeder, the aggregate voltage base was restored to 12.47 kV. Uniform synthetic ordinary-line sections were shortened from $(r, x) = (0.07, 0.01)\ \Omega$ to $(0.0175, 0.0025)\ \Omega$, while area-boundary sections use $(0.000025, 0.000025)\ \Omega$. Powers, DER capacities, voltage limits, loss terms, and SOC constraints were retained.

REFERENCES

[1] S. El Kazdadi, J. Carpentier, and J. Ponce, "Equality constrained differential dynamic programming," in *2021 IEEE International Conference on Robotics and Automation (ICRA)*, 2021, pp. 8053–8059.

[2] M. Xu, S. Gould, and I. Shames, "Line-search filter differential dynamic programming for optimal control with nonlinear equality constraints," 2026.

[3] A. R. Jha, S. Paul, and A. Dubey, "Analyzing the performance of linear and nonlinear multi-period optimal power flow models for active distribution networks," in *2025 IEEE North-East India International Energy Conversion Conference and Exhibition (NE-IECCE)*. IEEE, 2025, pp. 04–06.